

WEB

- **Servers and Websites:** The web consists of countless websites hosted on servers. Think of a server as a powerful computer that stores and serves web content. Each website is like a digital book or document.
- **Web Addresses (URLs):** Every website has a unique web address called a URL (Uniform Resource Locator). URLs are like the addresses of specific books in a library. For example, "<https://www.example.com>" is a URL.
- **Request and Response:** When you want to visit a website, you open a web browser (like Chrome, Firefox, or Safari) and type or click on a URL. Your browser then sends a request to the server associated with that URL, asking for the website's content.
- **Server Processing:** The server receives your request, locates the requested web page, and processes any necessary data (like retrieving content from a database). It then generates a response containing the web page's content.
- **Web Page Display:** Once the server has the response ready, it sends it back to your web browser. Your browser receives the response and displays the web page on your computer or device. This page can contain text, images, videos, links, and more.
- **Hyperlinks:** Within the web page, there are often clickable links (hyperlinks) that lead to other web pages. Clicking on a link sends a new request to the server for that specific page, and the process repeats.
- **Interactivity:** Some web pages have forms, buttons, or interactive elements. When you interact with these elements (e.g., submitting a form or playing a game), your browser sends data back to the server, which processes your input and responds accordingly.
- **Back and Forth:** As you navigate the web, you keep requesting new web pages, and your browser keeps fetching and displaying them. This back-and-forth communication happens seamlessly and quickly.
- **Security:** Web browsers and servers use security measures like HTTPS to ensure that your data is encrypted and secure during transmission. This helps protect your privacy and sensitive information.

DNS

DNS, or Domain Name System, is a fundamental technology used on the internet to translate human-friendly domain names (like www.example.com) into the numerical IP (Internet Protocol) addresses that computers use to identify each other on a network. DNS serves as a kind of "internet phone book" that helps your computer find the IP address associated with a domain name.

Here's how DNS works briefly:

- **Domain Name Resolution Request:** When you enter a website's URL (e.g., www.example.com) in your web browser or click on a link, your device sends a DNS query to a DNS resolver. The resolver is often provided by your internet service provider (ISP) or configured in your network settings.
- **Local Cache Check:** The DNS resolver first checks its local cache to see if it already knows the IP address associated with the domain you're trying to reach. If it has the information, it can provide the IP address immediately, speeding up the process.
- **Recursive Query:** If the resolver doesn't have the IP address in its cache, it becomes a DNS detective. It begins by asking the root DNS servers (the top-level authorities for the internet) for information about the domain. The root DNS servers point the resolver to the appropriate top-level domain (TLD) server.
- **TLD Server:** The TLD server, based on the domain's extension (e.g., .com, .org, .net), directs the resolver to the authoritative DNS server responsible for that specific domain.
- **Authoritative DNS Server:** The authoritative DNS server for the domain holds the authoritative information for that domain. It provides the resolver with the IP address corresponding to the requested domain.
- **Response to Resolver:** The authoritative DNS server sends the IP address back to the resolver, which caches this information locally for future use and forwards the IP address to your device.
- **Website Access:** Armed with the IP address, your device can now send an HTTP request to the web server associated with that IP address. The web server then sends back the web page content to your browser, which displays it on your screen.

In essence, DNS is the system that enables us to use user-friendly domain names instead of having to remember numerical IP addresses. It plays a crucial role in the functioning of the internet by ensuring that requests to access websites are directed to the correct servers, making web navigation easier and more efficient.